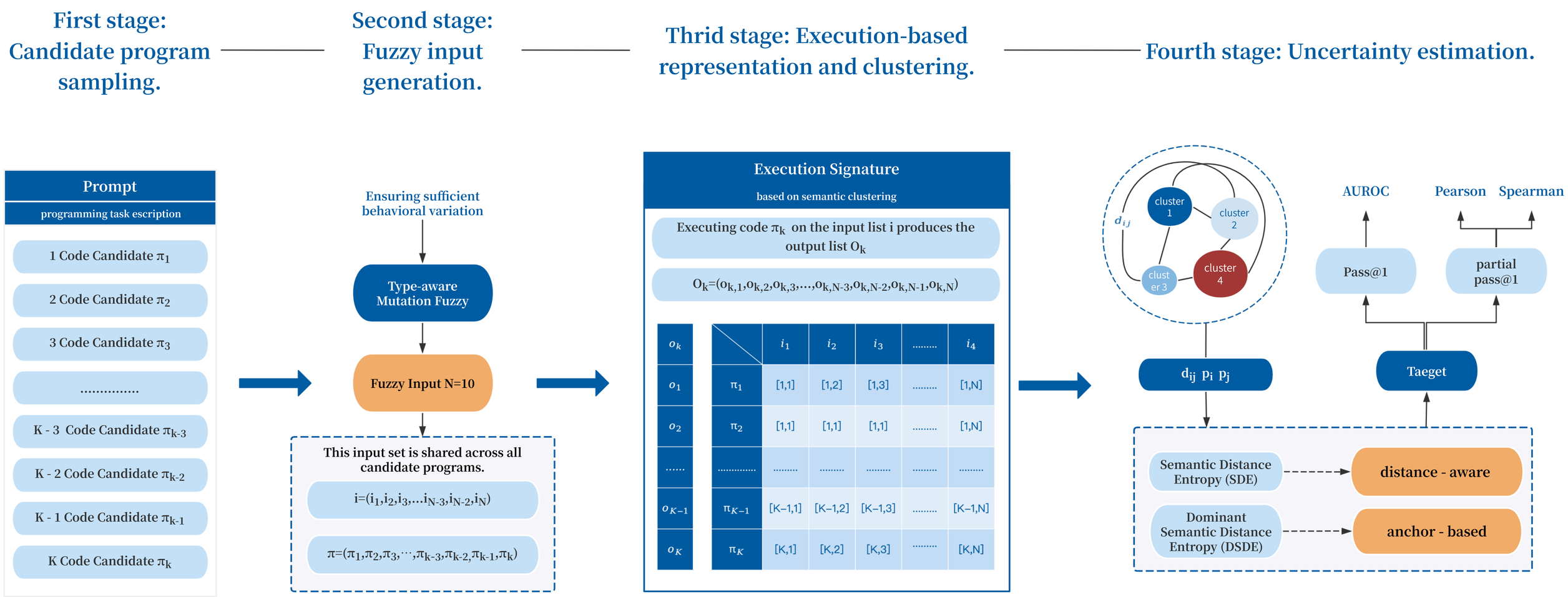
First stage: Candidate program sampling.
Second stage: Fuzzy input generation.
Thrid stage: Execution-based representation and clustering.
Fourth stage: Uncertainty estimation.
Prompt
programming task escription
1 Code Candidate π_1
2 Code Candidate π_2
3 Code Candidate π_3
.....
K - 3 Code Candidate π_{k-3}
K - 2 Code Candidate π_{k-2}
K - 1 Code Candidate π_{k-1}
K Code Candidate π_k
Ensuring sufficient behavioral variation
Type-aware Mutation Fuzzy
Fuzzy Input N=10
This input set is shared across all candidate programs.
$i=(i_1,i_2,i_3,\ldots i_{N-3},i_{N-2},i_N)$
$\pi=(\pi_1,\pi_2,\pi_3,\cdots,\pi_{k-3},\pi_{k-2},\pi_{k-1},\pi_k)$
Execution Signature
based on semantic clustering
Executing code π_k on the input list i produces the output list O_k
$O_k=(o_{k,1},o_{k,2},o_{k,3},\ldots,o_{k,N-3},o_{k,N-2},o_{k,N-1},o_{k,N})$
o_k | i_1 | i_2 | i_3 | ..... | i_4
o_1 | π_1 | [1,1] | [1,2] | [1,3] | ..... | [1,N]
o_2 | π_2 | [1,1] | [1,1] | [1,1] | ..... | [1,N]
..... | ..... | ..... | ..... | ..... | ..... | .....
o_{K-1} | π_{K-1} | [K−1,1] | [K−1,2] | [K−1,3] | ..... | [K−1,N]
o_K | π_K | [K,1] | [K,2] | [K,3] | ..... | [K,N]
d_{ij}
cluster 1
cluster 2
clust er 3
cluster 4
AUROC
Pearson
Spearman
Pass@1
partial pass@1
d_{ij} p_i p_j
Taeget
Semantic Distance Entropy (SDE)
distance - aware
Dominant Semantic Distance Entropy (DSDE)
anchor - based